TCM-NAN(λ) Creation Mechanism: World Emerges from 0 → 1 → All Things

Abstract

How does the universe evolve from absolute nothingness into a rich tapestry of phenomena?

TCM-NAN(λ) provides an extremely simple yet profound chain:

Difference ⇒ Amplification ⇒ Nonlinear Superposition ⇒ Q ⇒ Energy Manifestation ⇒ All Things

Where:

- λ = refresh delay field (related to parity violation of Yang-Mills and Lee)
- Asymmetric domain of λ: from −0 to 1
- Asymmetry spontaneously generates local structures (λball)
- Superposition of two λballs yields 1+1=3 (the third term $\delta\lambda_{12}$)
- Macroscopic geometric differences (velocity, angle, distance accumulation) amplify nonlinearity of angle differences
- Finally, the Q source term manifests "zero-point energy" as observable energy

1. Introduction

Traditional physics starts from matter: particles, fields, energy, conservation laws.

TCM-NAN takes a fundamentally different view:

The universe begins not with matter, but with difference.

- Where does difference come from?
- How is difference amplified?
- How does difference become energy?
- How does energy become structure?
- How does structure become everything?

The answer starts at the zero-point world of absolute nothingness.

2. Zero: No Volume, No Difference, No Energy — Not Even "Zero" Itself

2.1 Absolute Nothingness Cannot Sustain Its Own Existence

Absolute nothingness cannot allow itself to exist.

Mathematical expression:

$$\frac{dV_{\text{out}}}{d(\text{Nothing})} \to -\infty \tag{1}$$

Meaning:

- "Nothing" is not a stable state
- "Nothing" collapses due to its own contradiction
- It must jump from nothing to something

This is the starting point of the universe: 0.

Property	Zero-Point State (t=0)	Physical Picture
Time t	t=0	First refresh completed
Space λ	λ→∞	Infinite refresh delay, no correlation between points
Energy E	E=0	No difference, no accumulation
Information	Identical	All points are exactly the same

2.2 Essence of the Zero-Point World

The zero-point world is not empty space. It is the state just after the first refresh:

Property	Value	Meaning
Time	t=0	First refresh finished
λ	λ→∞	Infinite delay, no connection between points
Energy	E=0	No difference, no energy
Information	Identical	All points identical

Summary:

Zero-point world = identical world with no volume, direction, velocity, or difference.

Genuine zero has no volume — or the entire world is zero. Volume is infinite. This infinite zero cannot maintain perfect global uniformity; it collapses into −0 ↔ +0 oscillations.

Due to asymmetry, it cannot return to pure zero.

Thus "1 (existence)" is born.

2.3 Arithmetic Relation Between λ and Time

$$\lambda = \frac{t_1}{t} \tag{2}$$

When $t \to 0^+$: $\lambda \to \infty$

- The closer time approaches zero, the larger λ becomes
- The more "disconnected" space is
- The fewer differences exist

Differences in position, time, frequency, and phase of zero break symmetry — no energy input, origin of non-conservation.

3. From Zero to λ: Refresh Initiates, Difference Emerges

3.1 Refresh Sequence: λ Decreases with Time

When $t > 0$, refreshing begins:

$$\lambda_1 = \Delta t_1,\ \lambda_2 = 2\Delta t_1,\ \lambda_3 = 3\Delta t_1, \ldots \tag{3}$$

Abstract form:

$$\lambda(t) \searrow 1 \tag{4}$$

Meaning:

- Refresh delay gradually decreases
- Connections form between spatial points
- Differences begin to appear
- Difference is the prerequisite for energy generation

3.2 Evolution of Cosmic λ

Stage	Time	State
Zero Point	t=0	λ=∞, fully identical
Early	t→0⊠	λ very large, initial differences
Present	13.8 Gyr	λ≈1, significant differences
Future	t→∞	λ→0, fully synchronized

Cosmic evolution path:

Extremely disconnected (λ→∞) → Finite difference (λ≈1) → Fully synchronized (λ→0)

4. Asymmetry of λ: −0 → 1, Seed of Creation

4.1 Asymmetric Domain of λ

$$\lambda \in [-0,\ 1] \tag{5}$$

Only −0 exists on the negative side; the positive side is a full continuous interval.

4.2 Asymmetric Potential

$$V(\lambda) = \frac{1}{2}m^2\lambda^2 + \frac{g_3}{3}\lambda^3 + \frac{g_4}{4}\lambda^4 + \cdots \tag{6}$$

Constraint:

$$\lambda \geq -0 \tag{7}$$

Under this constraint, the λ field spontaneously forms:

- Localized structures
- Solitons
- Stable modes

5. λball: Prototype of Matter (No Volume, No Hardness, Pure Field)

$$\lambda(r,t) = \lambda_0 + \delta\lambda(r,t) \tag{8}$$

A λball is not a physical sphere, but a stable standing wave of the λ field in an asymmetric potential.

Intrinsic properties:

- Locality
- Stability
- Gradient energy (corresponds to mass, gravity)
- Nonlinear superposition (corresponds to interaction)

Conclusion: Matter = stable mode of λ

6. 1+1=3: Nonlinear Superposition of Two λballs

6.1 Total Field Is Not Simple Linear Superposition

$$\lambda = \lambda_0 + \delta\lambda_1 + \delta\lambda_2 + \delta\lambda_{12} \tag{9}$$

- $\delta\lambda_1$: first λball
- $\delta\lambda_2$: second λball
- $\delta\lambda_{12}$: third term (core of nonlinearity)

6.2 Origin of the Third Term $\delta\lambda_{12}$

Nonlinear potential breaks superposition:

$$V(\lambda_1 + \lambda_2) \neq V(\lambda_1) + V(\lambda_2) \tag{10}$$

Cross term generates the third component:

$$\delta\lambda_{12} \propto |\nabla\delta\lambda_1 \times \nabla\delta\lambda_2| \tag{11}$$

6.3 Angle Difference Triggers the Third Term

Gradient direction difference:

$$\angle(\nabla\delta\lambda_1, \nabla\delta\lambda_2) = \Delta\theta \tag{12}$$

Interaction:

$$\delta\lambda_{12} \sim \sin\Delta\theta \tag{13}$$

- Parallel (Δθ=0): no third term, no energy
- Angular difference (Δθ>0): third term appears, energy manifests

7. Q Source Term: Energy Manifests from the λ Field

$$Q = \alpha(\nabla\lambda)\rho_m \tag{14}$$

Substitute superposition:

$$Q \approx \alpha\left(\nabla\delta\lambda_{12}\right)\rho_m \tag{15}$$

Energy density:

$$\rho_E \propto Q^2 \tag{16}$$

Core conclusion: Energy comes not from collision, but from nonlinear superposition of the λ field.

8. Macroscopic Geometry: Velocity Difference, Angle Difference, Distance Accumulation

8.1 Angle Difference (Grows Linearly with Time)

$$\Delta\theta(t) = v\left(\frac{1}{R_1} - \frac{1}{R_2}\right)t \tag{17}$$

8.2 Arc-Length Difference (Geometric Amplification)

$$\Delta x(t) = v\left(\frac{R_2}{R_1} - 1\right)t \tag{18}$$

Amplification factor:

$$k = \frac{R_2}{R_1} \tag{19}$$

Simplified:

$$\Delta x(t) = v(k-1)t \tag{20}$$

8.3 Quadratic Amplification of Energy-Like Quantity

Instantaneous effective:

$$E_{\text{eff}} = \frac{1}{2}v^2(k-1)^2 \tag{21}$$

Cumulative effective:

$$E_{\text{acc}}(t) = \frac{1}{2}v^2(k-1)^2\left(\frac{t}{t_0}\right)^2 \tag{22}$$

9. Macroscopic Geometry → Microscopic λ Field: Complete Creation Chain

$$\Delta v, \Delta\theta \Rightarrow \Delta x(t) \Rightarrow \text{Gradient Direction Difference} \Rightarrow \delta\lambda_{12} \Rightarrow Q \Rightarrow E \tag{23}$$

Summary: Triggered by macroscopic difference, executed by microscopic λ field.

10. From Zero-Point Energy to All Things: TCM-NAN Cosmic Generation Formula

Core principle:

Differences in velocity, angle, direction, phase continuously create new energy

These geometric properties require no energy input — they are the essence of non-conservation and creation from nothing.

Macroscopic Conclusions

- Angle difference: grows linearly in time
- Arc-length difference: grows linearly, amplified by radius ratio
- Energy-like quantity: quadratic amplification
- Provides a geometric lever arm for the λ field

Mathematics:

$$E_{\text{create}}(t) = E_0 \cdot F\big(\Delta v(t), \Delta\theta(t), \Delta\phi(t)\big) \tag{24}$$

Exponential amplification:

$$\Delta X(t) = \Delta X_0 e^{Ht} \tag{25}$$

Final source term:

$$Q = \alpha(\nabla\lambda)\rho_m \tag{26}$$

Complete Creation Steps

1. Zero-point (nothing): $\lambda=\infty$, identical, no volume, no difference
2. λ field (background): $\lambda\in[-0,1]$, asymmetric potential, refresh delay
3. λ-ball (particle): localized structure of the field
4. 1+1=3 (interaction): angle difference generates third term $\delta\lambda_{12}$
5. Q source (energy): gradient $\nabla\delta\lambda_{12}$ activates Q, releases energy
6. Velocity-difference amplification: angle accumulation, linear/exponential energy boost

11. Full Creation Path

1. Absolute nothing → ontological pressure → zero-point energy
2. Zero-point world: t=0, $\lambda=\infty$, identical, no difference
3. Refresh starts: λ decreases, difference appears
4. Asymmetry of λ: $-0 \to 1$
5. λ-ball: localized stable mode = prototype of matter
6. Two λ-balls overlap → third term $\delta\lambda_{12}$
7. Macroscopic geometry: velocity difference, angle difference, distance accumulation
8. Microscopic λ field: gradient direction difference → Q source
9. Energy manifests → structure → galaxies → life → you

12. Core Insights of TCM-NAN

- The universe begins with difference, not matter
- Difference arises from λ-asymmetry, which arises from logical instability of the zero-point world
- Macroscopic geometric differences are projections of nonlinear λ-field superposition
- The microscopic λ field ($\delta\lambda_{12}$) is the true origin of energy creation

Ultimate Conclusion:

- All things = stable modes of λ
- Energy = release of zero-point energy
- The universe = history of difference amplification

Core Stage Summary Table

Stage	Description	Key Formula
0. Origin	Self-negation of absolute nothing	$\frac{dV}{d(\text{Nothing})} \to -\infty$
1. Background	Refresh delay of λ field	$\lambda(t) \in [-0, 1]$
2. Structure	λ-ball (matter prototype)	$\lambda(r, t) = \lambda_0 + \delta\lambda$
3. Collision	Angle difference creates third term	$\delta\lambda_{12} \propto (\nabla\delta\lambda_1 \times \nabla\delta\lambda_2)$
4. Energy	Q source manifestation	$Q = \alpha(\nabla\lambda)\rho_m$
5. Amplification	Velocity → angle → energy	$\Delta\theta(t) = \Delta\theta_0 + \Delta\omega, t$

Appendix A: Complete Formula List

A.1 Zero-Point World & Basic λ Relations

No.	Formula	Meaning
A1	$\frac{dV}{d(\text{Nothing})} \to -\infty$	Nothing cannot sustain itself
A2	$\lambda = \frac{t1}{t}$	λ inversely proportional to time
A3	$t \to 0^+ \Rightarrow \lambda \to \infty$	Infinite delay at zero point
A4	$\lambda \in [-0, 1]$	Asymmetric domain of λ

A.2 λ Field Dynamics & Potential

No.	Formula	Meaning

A5	$V(\lambda) = \frac{1}{2}m^2\lambda^2 + \frac{g3}{3}\lambda^3 + \cdots$	Asymmetric potential
A6	$\lambda \geq -0$	Hard wall on negative side
A7	$\lambda = \lambda_0 + \delta\lambda$	Localized form of λball

A.3 Nonlinear Superposition (1+1=3)

No.	Formula	Meaning
A8	$\lambda = \lambda_0 + \delta\lambda_1 + \delta\lambda_2 + \delta\lambda_{12}$	Total field superposition
A9	(\delta\lambda_{12}\propto	\nabla\delta\lambda_1\times\nabla\delta\lambda_2
A10	$\angle(\nabla\delta\lambda_1, \nabla\delta\lambda_2) = \Delta\theta$	Gradient direction difference

A.4 Q Source & Energy Manifestation

No.	Formula	Meaning
A11	$Q = \alpha(\nabla\lambda)\rho_m$	Creation source term
A12	$Q \approx \alpha(\nabla\delta\lambda_{12})\rho_m$	Third term dominates Q
A13	$\rho_E \propto Q^2$	Energy density

A.5 Macroscopic Geometric Relations

No.	Formula	Meaning
A14	$\Delta\theta(t) = v\left(\frac{1}{R_1} - \frac{1}{R_2}\right)t$	Angle difference
A15	$\Delta x(t) = v\left(\frac{R_2}{R_1} - 1\right)t$	Arc-length difference
A16	$k = \frac{R_2}{R_1}$	Amplification factor

A.6 Energy-Like Quantities

No.	Formula	Meaning

A17	$E_{\text{eff}} = \frac{1}{2} v^2 (k-1)^2$	Instantaneous energy-like quantity
A18	$E_{\text{acc}}(t) = \frac{1}{2} v^2 (k-1)^2 \left(\frac{t}{t0}\right)^2$	Cumulative energy-like quantity